

Participant Information Sheet (PIS) Template

MultiAge: Language production in multilingual older adults living with cognitive impairment

Invitation and brief summary

We would like to invite you to take part in our research study, led by researchers at the University of York. The project lead is Dr Angela de Bruin, a Senior Lecturer in the Department of Psychology at the University of York.

In this study we look at how multilinguals use their languages. A multilingual is someone who can communicate in more than one language. We are especially interested in people who speak your language(s) and English. We are looking at how language differs between multilinguals living with or without cognitive impairment. People living with mild levels of cognitive impairment have problems with their memory and thinking that continue for more than a few months. Their memory and thinking are not working as well as expected for the person's age and background. These problems can include difficulties remembering recent events or learning new information, struggling to make decisions or work through everyday problems, finding it more difficult to focus on tasks, and/or having difficulties finding the right word when speaking. We here define "mild" as meaning that although the symptoms may be troubling, the person is still able to do most everyday activities.

Language changes can tell us something about changes in how people function. However, what we know about these language changes so far is based on research with people who speak only one language. In this study, we want to understand better how language changes in *multilinguals* who are living with cognitive impairment.

Participating in this study is entirely optional. Your decision to take part (or not) will not influence the care you receive. You can also stop the study at any time. If you change your mind after taking part, you can contact us until 31 March 2028. We will then remove all data about you.

The study will take approximately 2 to 2.5 hours. We will do this in a location of your choice. We will explain each part of the task to you and will do everything together. In the main task, we will show you simple pictures. We then ask you to describe each picture using English and/or your language. We will also ask you some questions about your health and language background. We use a short interview to learn more about how you use your languages. Finally, we will do a few short tests to look at how well you remember new information, respond to questions, and can pay attention to different things. If you agree, we will also ask one of your family members or friends to answer a few questions about you.

You receive £25 and a few small gifts for taking part.

If you are willing to, we may ask you to wear a textile and plastic cap during the main task. With this cap, we can look at how active your brain is during the task. This cap is called "fNIRS" and used widely around the world with adults, children, and even babies. It is completely safe and uses infrared light that shines on your head.

Wearing the cap is completely optional, and you can also take part without wearing it.

We are inviting you to take part if:

- You are between 60 and 75 years old
- You can have a conversation in Mirpuri-Pahari-Pothwari, and/or Punjabi/Panjabi, *and* English.
- You are living with cognitive impairment
- As a child, you did not have any language or reading difficulties such as stuttering or dyslexia
- You can see information on a computer screen and hear spoken information

Before you decide to take part, please read through this information and watch a [short video in your language](https://www.youtube.com/watch?v=BGiShluXGjo) (<https://www.youtube.com/watch?v=BGiShluXGjo>). One of our multilingual team members will also go through this information with you and answer any questions you have. **Please do get in touch if anything is unclear or you would like more information in your language.** Our contact details are:

University of York project team: Dr Angela de Bruin, Dr Zlatomira Ilchovska, Mr Imran Arif

Website: <https://multiage.github.io/>

Email address: multiage-project@york.ac.uk

Phone number: 07435017504

In the study, we will ask you for some information about yourself. This will include:

- your contact details
- information about your health and personal background
- audio recordings while you do the language production tasks
- with your permission (where relevant), we will also ask your local care team for some relevant information from your medical records
- If you choose to wear the fNIRS cap, we also ask to take a short video of your head while wearing the cap.

We will only use the information that we need for the research study. We will let very few people know your name or contact details, and only if they really need it for this study. Everyone involved in this study will keep your personal data safe and secure. We will follow all privacy rules. The audio recordings we make will only be accessible to the project team at the University of York. They are saved on secure University of York file storage. We will transcribe what you said and then delete the recordings. The responses you give during the study will be saved with a random number instead of your name. In the reports we write, no-one can work out who you are. At the end of the study, we will save some of the data, in case we need to check it, for future research, and in line with the University of York's retention period. With your consent, we keep your contact details if you would like to receive a short summary of the study findings and/or invitations for future studies.

MultiAge and the researchers working on it want to make science better and more transparent. To do this, we are planning to share the study data publicly. We will only share data that other people outside the project team cannot link back to you.

You will not receive individual feedback on any of the tasks. However, if you are interested to hear more about this study and the results, we can send you a study report after study completion.

We will make sure no-one can work out who you are from the reports we write.

The information pack tells you more about this.

What is involved in this study?

The background and purpose of the research

Multilinguals (people who speak more than one language) can speak words in one or multiple languages. Sometimes multilinguals use different languages within a conversation or even within a sentence. We call this “language switching”. In this study, we want to know more about how language production and language switching differ between multilinguals who are living with cognitive impairment and multilinguals of similar ages without cognitive impairment.

Language production can change with ageing. Language changes can also be an early indication of cognitive impairment. Most of what we know, however, is based on people who speak one language. In this study, we want to understand better how language changes in multilinguals like yourself. In the future, we hope this work can help us to develop new assessments of cognitive functioning that are more suitable for multilinguals.

We are working with multilinguals who speak Mirpuri-Pahari-Pothwari and/or Punjabi/Panjabi (in addition to English), as these languages are very commonly spoken across the North of England and the Midlands. We are hoping to work with 30 people living with cognitive impairment in this study. We will compare these data to people of the same ages who do not have cognitive impairment.

If you are interested in taking part, please find more information about the study below. You can also find a [short information video in your language](https://www.youtube.com/watch?v=BGiShluXGjo) (<https://www.youtube.com/watch?v=BGiShluXGjo>).

Please call our multilingual project coordinator (07435017504) if you would like to receive more information in your language.

What would taking part involve?

This project is a research study. Your decision to take part in the study does not influence the care you receive.

If you choose to take part, the study will include the following steps:

- We will make an appointment with you. The study will take place in a location of your choice, for example your home or a nearby space (this can be a library or community centre).
- You will do the study with one or two researchers from the University of York (Mr Imran Arif and Dr Zlati Ilchovska). At least one person speaks your language(s).
- We will talk through this information sheet and the study with you. If you are happy to take part, we will ask you to complete a consent form.

You will then complete the following tasks:

- Scene description task (language production task). You will see a simple picture on a computer screen, for example a girl eating spaghetti in a kitchen. We will ask you to describe the picture in one sentence. Sometimes we will ask you to use English and sometimes your language. Sometimes you can choose. You do not have to work with the computer yourself. The study leader will help you. Your responses will be audio recorded.
- Short interview. We will ask you some simple questions to look at how you speak your languages. Some questions need to be answered in English and some in your language. Your responses will be audio recorded.
- Questionnaire. The study leader will ask you questions about your language background, for example how often you use each language. We will also ask you a few other questions about your health and personal background.
- A short task asking you to watch a video of different rooms in a house. We will then also ask you some questions about the rooms you saw.
- A few questions to better understand your overall cognitive functioning. This will look at how you respond to simple spoken questions, language production, memory, and attention.

You will not receive feedback on any of the tasks. We will also ask one of your friends or family members to answer a few questions about you. This will ask them if they have noticed changes in your memory and behaviour over the past ten years. They can do this if they are there during your study visit. Otherwise, please share an information sheet with them. They can then contact us directly.

During the language production tasks, we will audio record your spoken responses. We make these recordings so that we can find the information we need to answer our research questions, for example how often you use different languages. We will make a transcription of what you said during the recordings. These transcriptions will be made by one of the members of the project team and not by external parties. If you include any information that can identify who you or other people are, we will remove this from the transcription. These recordings will be directly recorded to a University of York managed laptop and are stored in the project leader's secure University of York file storage. The audio recordings are only accessible to the project team at the University of York and will not be shared with any third parties. They will be deleted by the project lead once we have finished processing the recordings.

We will also ask you if you would be willing to wear a cap during a part of the study. This cap can measure how active your brain is during different parts of the task. This is non-invasive, meaning that

it does not do anything to or change anything in your brain. You can find more information about this cap in the attached fNIRS information sheet. It is entirely your choice if you want to wear this. You can still take part in the study if you do not want to. You can also first see what the cap looks like in person and then decide if you are comfortable wearing it. You are free to change your mind about this at any time.

You do not need to prepare anything before the study visit. If you would like to wear the fNIRS cap, please make sure your hair is dry and clean. We will walk you through every step of the study and help you with all parts. The study will take about 2-2.5 hours to complete. This includes breaks and time for questions. You can take breaks whenever and for as long as you prefer. You will receive £25 and a few small gifts for taking part.

Please note that if, in the course of our study, we discover something that raises concerns about your safety or the safety of others, we are obliged to seek further expert help and advice.

Who can take part in the research?

You have been invited to take part because you speak English and Mirpuri-Pahari-Pothwari and/or Panjabi/Punjabi.

Please ONLY take part if:

Age: You are between 60 and 75 years old. Please contact us if you are a few years younger or older than this age group and would like to take part.

Language:

- You speak English and Mirpuri-Pahari-Pothwari and/or Panjabi/Punjabi. By “speak” we mean that you can have a conversation in each language.
- You use both languages at least sometimes in your daily life (you do not ONLY use English OR Mirpuri-Pahari-Pothwari and/or Panjabi/Punjabi in your daily life).

Health:

- You are living with cognitive impairment
- You did not have any language or reading disorders as a child (e.g., stuttering or dyslexia)
- You can see simple pictures on a computer screen. It is fine if you wear glasses or contact lenses.
- You can hear spoken information without difficulties. It is fine if you wear a hearing aid.

Do I have to take part?

No, participation is optional. Please read through this information sheet and discuss any questions you might have with a member of the research team. We will give you a copy of the information sheet to keep. If you decide to take part, you will be asked to complete a participant consent form. If

you change your mind at any point during the study, you can stop without having to provide a reason. You can simply tell the research leader to stop.

If you would like to remove your data at any point before or after study completion, you can do this by contacting Dr Angela de Bruin at angela.debruin@york.ac.uk or by calling 07435017504. We might not always be able to do this if it means we cannot use your data to do the research. If so, we will tell you why we cannot do this. ***For this project, you can ask us to remove your data until 31 March 2028. After this date, we will start analysing the data and remove any connection between your data and name. From that point, we can therefore no longer identify which information belongs to you.***

What are the possible benefits of taking part?

There are no immediate personal benefits to taking part. This is the first time we are studying language production in your community. We therefore cannot give you individual feedback on your performance. This is a first research project to see how language might change in multilinguals living with cognitive impairment. We hope to use this research to find new ways to use language changes to tell us if someone might show signs of cognitive impairment. This can provide future benefits to multilingual communities like yours.

Recompense

To recompense you for taking part in this research, you will receive £25 and a few small gifts (tote bag and brain “stress ball”). If you have to travel to the testing location, we will cover your travel time and travel expenses (an additional £10 per 30 minutes). To reimburse participants, we are required to ask for their home address and name to meet HMRC requirements. Your data (name and home address) will be shared with the administration team in the Department of Psychology so that they can process the reimbursement. In line with HMRC retention requirements, these data will be kept for seven years. Your data will not be shared with others outside the people who need access to this information and will not be used for any other purposes. They will also be saved separately from the research data collected in this study. Please be aware that if you are paid more than £1000 over the course of a year, you will need to complete a Self-Assessment form for HMRC.

What are the possible disadvantages and risks of taking part?

There are no immediate disadvantages or risks of taking part. This study will not influence the care you receive. You might find the study tiring. The study is relatively long (2-2.5 hours) and involves looking at a computer screen. You can take breaks whenever you prefer. You can also ask to stop the study at any time. If you feel tired and prefer to complete the study another time, we can also do the study in multiple shorter parts on different days.

If you choose to wear the cap to look at your brain activation (fNIRS, please see the additional information sheet about this), you might experience slight discomfort. For example, you might find the cap feeling a little tight or warm. If you experience any discomfort, please let us know

and we can make adjustments or you can stop wearing the cap. You can still take part in the study without wearing the cap.

What will happen with my data?

Please see the section “How will we process your data in line with UK GDPR” below for detailed information.

Under article 6 of the UK General Data Protection Regulation (UK GDPR), the lawful bases we rely on for processing personal data are:

- For the performance of a task in the public interest or in the exercise of official authority vested in the controller

We will use your data to answer our research questions, looking at changes in language production when living with cognitive impairment. The results of this study will be presented in verbal outputs (presentations) and written outputs (reports and academic publications). These outputs will not be able to identify you. We will also make a summary of the results available on our website (<https://multiage.github.io/>). If you give consent below, we will send you a copy of the results summary.

Your data will be saved on one of the University of York’s approved, secure storage options. This can include use of Google Drive (the University of York has a GDPR compliant agreement in place) and other software approved by the University of York.

Our research group is committed to open research practices, which includes sharing research data through public repositories such as the Open Science Framework. This way, the data you provide during the study tasks will be shared publicly with others. We will only share data that cannot identify who you are to people outside the research team. We will never share any personal information like your name or contact details publicly. We will not share the audio recordings either. The information we do share is given in a format that keeps your identity confidential while still being very useful to researchers and clinicians. The data we share can be accessed by other people who might use them for other research questions that are not known to us. The research team and others might therefore use these data for secondary research and/or teaching purposes.

If you have indicated you would like to receive a copy of the results summary, we will keep your contact information until we send you the final study report. If you would like to receive invitations for future studies, we will also save your contact details for five years. If you would like us to remove your information from this list at any point, please contact the project team (please see the contact details at the start or end). With your consent, we might also share your contact details with collaborators at the University of Sheffield, so that they can send you a study invitation about another project looking at multilingualism and cognitive impairment. You are under no obligation to respond to any future study invitations.

Who is funding the study?

This project was selected for an ERC starting grant by the European Research Council as part of Horizon Europe. The funding for the project is provided by the UKRI as part of the Horizon Europe

guarantee scheme for selected ERC grants (Grant Number EP/Y036522/1). UKRI (UK Research and Innovation) is the public body of the UK Government that directs research and innovation funding.

Who has reviewed the study?

This study has been reviewed by an independent group of people, called a Research Ethics Committee to protect your safety, rights, wellbeing, and dignity. This group includes both NHS staff and members of the public. This study has been reviewed and given favourable opinion by the NHS East Midlands - Derby Research Ethics Committee.

Questions

If you have any questions now or in the future, please contact:

University of York project team: Dr Angela de Bruin, Dr Zlatomira Ilchovska, Mr Imran Arif

Website: <https://multiage.github.io/>

Email address: multiage-project@york.ac.uk

Phone number: 07435017504

If you have any concerns or complaints about how your data are used, please contact: the University of York's data protection officer: dataprotection@york.ac.uk

Data Protection Officer, University of York, Heslington, York, YO10 5DD.

If you wish to make a complaint about the study to someone independent of the project from the University of York, please contact our local ethics committee (psyc529@york.ac.uk)

How will we process your data in line with UK General Data Protection Regulation (UK GDPR)?

How will we use information about you?

We will need to use information from you as well as some information from your medical record for this research project. This information will include:

- Your name and contact details. This information is collected so that we can contact you and schedule the appointment. We will also use this to link your study responses to your friend's or family member's responses to their short survey. Unless you ask us to keep your information, we will remove this information at the end of the data collection period (scheduled to be March 2028).

- Your contact details if you would like to receive a summary of the study's findings, after we have finished collecting and analysing the data. We delete this information after sending you the final study report.
- Your contact details if you would like to receive invitations for future studies. We would keep this information for five years. You can contact us at any time to have your information removed.
- Your NHS number (only if you received the study invitation from your clinical team). This is so that we can ask your clinical team to share relevant information from your medical records with us. The project team at the University of York *will not* have direct access to your medical records. We will only receive information that is directly relevant for our study. We remove your NHS number after having received the relevant information.
- Audio recordings, where we record your spoken responses during the language production tasks. We record your responses so that we can write down what you said and get the information we need to answer our research questions (for example, how often you switched languages). We delete these recordings when they are no longer needed for our study.
- If you choose to wear the cap measuring your brain activation (fNIRS), we also ask to take a short video of your head while wearing the cap. This is to help us when processing the data and to better understand how the cap was placed. We delete these recordings when they are no longer needed for our analyses.
- Language background questionnaire. This questionnaire will include some questions about you and your background, including (for example) your heritage region, some questions about your health, and questions about your languages. The questionnaire responses, together with your other study responses, will be kept by the project team for ten years after last access.
- Your name and address to process the reimbursement for taking part in this study. Please note that this information needs to be shared with our administration team at the University of York. They are required to hold this information for seven years to comply with HMRC regulations.
- Your name and signature on the consent form will be retained for ten years after the end of the study, after which they will be destroyed.

People will use this information to do the research or to check your records to make sure that the research is being done properly.

People who do not need to know who you are will not be able to see your name or contact details. Your data will have a code number instead.

The University of York is the sponsor of this research. **The University of York** is responsible for looking after your information. We will share your information related to this research project with the following types of organisations:

- Where needed for the study, information will be shared between the University of York team and the clinical collaborators on this project. People will use this information to do the research or to check your records to make sure that the research is being done properly.
- Where relevant, your name and/or NHS number will be shared with your clinical team so that the University of York project team can receive relevant information for this project about your medical background.
- If you give consent, we will also share your contact details with collaborators at the University of Sheffield, to receive a study invitation for their project on multilingualism and cognitive impairment. This is optional.

We will keep all information about you safe and secure by:

- We make sure only the study team and where relevant clinical team will have access to your name and contact details. The exceptions are: a) data we need to share with the admin team to process your reimbursement; b) if you give us consent to share your contact details with our collaborators at the University of Sheffield for future studies on multilingualism and cognitive impairment; c) if/where relevant, access is granted to a minimal number of individuals if necessary for quality control and auditing purposes.
- The responses you provide during the study will be saved with a random code. This makes sure your identity is not saved in the same file as any responses you provide during the study. We will keep a separate file linking this random code to your name until March 2028. After that date, we will start our analyses using your random code only.
- We do not ask for more personal information about you than we need for this study.
- All personal information will be stored on the University of York's secure storage systems. This can only be accessed by members of the study team. Where data need to be exchanged outside the University of York, secure transfer systems will be used (e.g., University of York's secure DropOff service).

International transfers

We may share or provide access to data about you outside the UK for research related purposes to:

- We intend to share research data publicly on data sharing platforms such as the Open Science Framework (OSF). This allows other researchers to check our data and analyses and to answer other research questions they might have. We will only share data that cannot identify who you are to people outside the project team. We will not share audio recordings or individual health-related information. Initially these data will be pseudonymous only, meaning that the University of York project team might still be able to identify you. The shared data will become fully anonymous after deletion of the raw research data by the project team.

If this happens, we will only share the data that is needed. We will also make sure you can't be identified from the data that is shared where possible. This may not be possible under certain circumstances – for instance, if you have a rare illness, it may still be possible to identify you. If your data is shared outside the UK, it will be with the following sorts of organisations:

- Your research data will be accessible to anyone with an internet connection, anywhere in the world. We will make sure the data we share does not allow other people to identify who you are. Other people can use the research data to answer new questions not known to us.

We will make sure your data is protected. Anyone who accesses your data outside the UK must do what we tell them so that your data has a similar level of protection as it does under UK law. We will make sure your data is safe outside the UK by doing the following:

- We will only share data outside the UK that cannot identify who you are to people outside our project team. We will also not share individual information that is health related. People will only be allowed to access your data if they do not attempt to (re-)identify who you are.

How will we use information about you after the study ends?

Once we have finished the study, we will keep some of the data so that we can check the results. We will write our reports in a way that no-one can work out that you took part in the study. We will keep your study data for a maximum of ten years after last access. The study data will then be fully anonymised and securely archived or destroyed.

What are your choices about how your information is used?

You can stop being part of the study at any time, without giving a reason, but we will keep information about you that we already have. You have the right to ask us to access, remove, change or delete data we hold about you for the purposes of the study. You can also object to our processing of your data. We might not always be able to do this if it means we cannot use your data to do the research. If so, we will tell you why we cannot do this.

Where can you find out more about how your information is used?

You can find out more about how we use your information, including the specific mechanism used by us when transferring your personal data out of the UK:

- our leaflet (<https://www.hra.nhs.uk/planning-and-improving-research/policies-standards-legislation/data-protection-and-information-governance/gdpr-guidance/templates/template-wording-for-generic-information-document/>) and University of York privacy notice for research participants (<https://www.york.ac.uk/records-management/dp/your-info/privacynotice-researchparticipants/>)
- by asking one of the research team
- by sending an email to multiage-project@york.ac.uk, or
- by ringing us on 07435017504.
- by contacting the University of York's data protection officer: dataprotection@york.ac.uk or Data Protection Officer, University of York, Heslington, York, YO10 5DD.